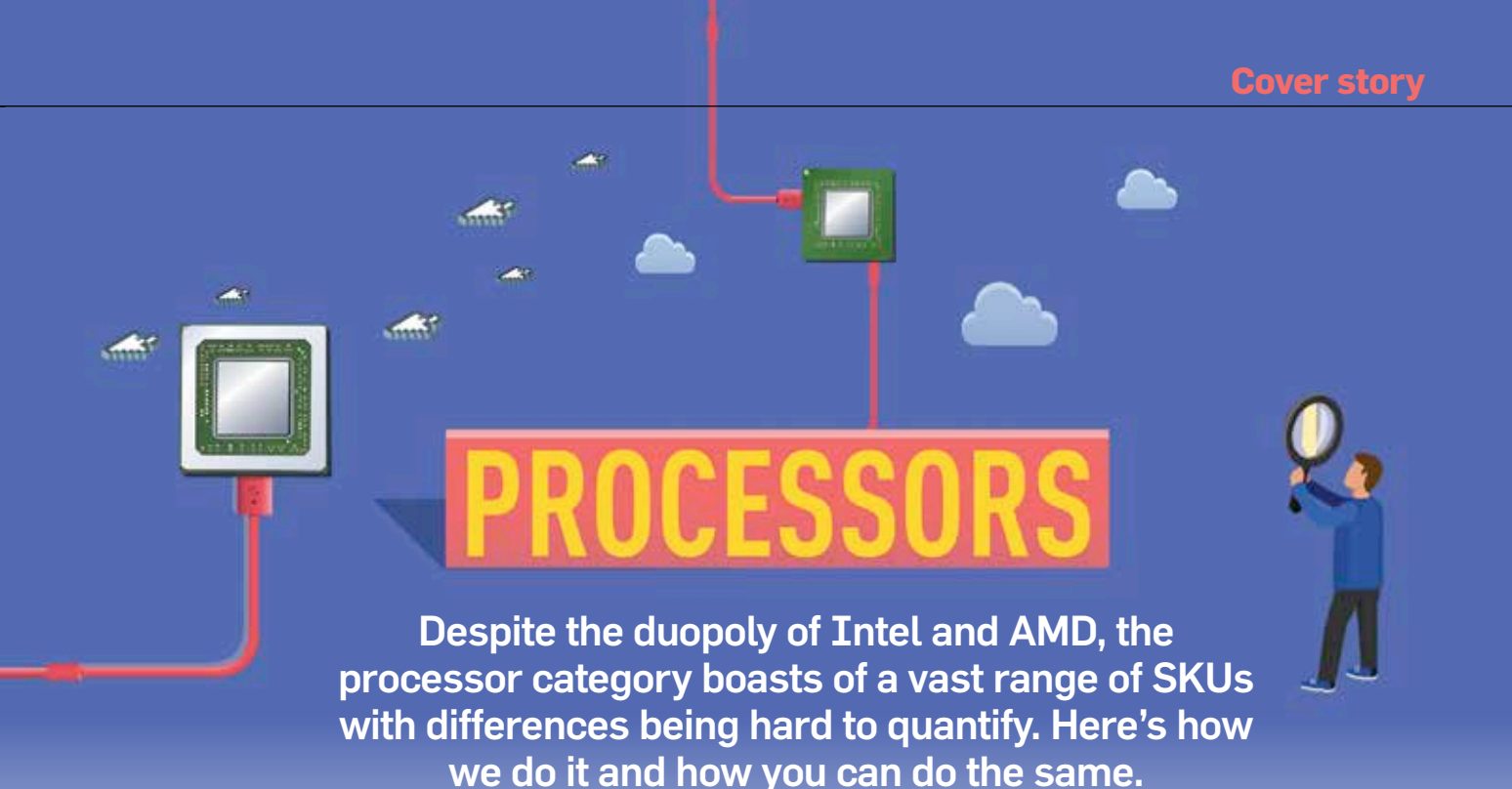




Despite the duopoly of Intel and AMD, the processor category boasts of a vast range of SKUs with differences being hard to quantify. Here's how we do it and how you can do the same.

By **Mithun Mohandas**

The heart of a computer is one that is shrouded behind so many veils making it mysterious as well as confusing. And depending on what you are looking at, the characteristics that matter are different. Intel and AMD are the only two players in the market and both have an awful number of SKUs in the market and the nomenclature that both follow can be said to be works of modern art. That is, interpreting the model numbers require a fair amount of research and effort which puts it out of the league of most people.

Types

Depending on what kind of device it has to power, the processor takes on different roles. There are processors that also run the task of rendering graphics and there are specialised processors which are suited for low-powered implementations. There are server grade processors as well but that's something we won't be touching upon given that most of our readers are not into them.

Desktop Processors

The run of the mill processors from Intel and AMD boast of pretty well rounded capabilities. Both manufacturers have plenty of options when it comes to how

many cores you'd want. Each manufacturer has segregated their offerings across three buckets – Extreme, Mid-range and Low-end. Marketing terminologies vary but the overall picture is the same. You can spend as low as ₹2,500 or sell both your kidneys and get a processor that cost upwards of ₹74,000.

Unless it's an enthusiast processor, there is bound to be a graphics processor on the processor die itself. There are exceptions to this rule. Both Intel and AMD have pretty good on-chip graphics processors. Intel's HD Graphics improved greatly thanks to a patent sharing agreement with NVIDIA and AMD's acquisition of ATI helped it in a similar manner.

On a general note, Intel's processors use labels like i7, i5 and i3 apart from Pen-

tium and Celeron. AMD processors are sorted into two series – FX for enthusiasts and APUs for all segments.

Mobile Processors

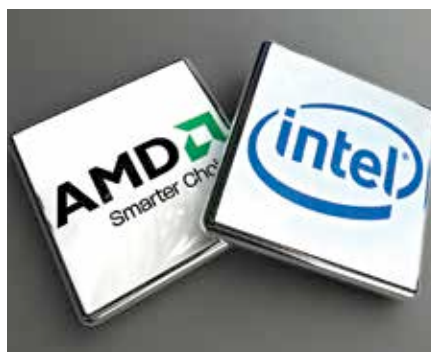
Laptops and even tablets these days incorporate processors which are toned down variants of what you'd find in your desktop PCs. This is necessitated due to the space constraints in these devices. Had these used desktop processors, then you'd pretty much be able to use your laptop as a computer-cum-electric-hot-plate.

SoCs

SoCs are what you tend to find in smart devices and of late, even Intel and AMD have forayed into a space that was dominated by ARM. SoCs used to be a few rungs below mobile processors but if we look at the past few years then they've hit a point of convergence where the two are nearly indistinguishable.

What do the numbers mean?

Processor benchmarks often give out scores that are a combination of multiple individual tests that have already been weighted and combined. These benchmarks are trace based, i.e. the work done by the processor under different scenarios



An age old battle between bitter rivals